Improving Customer Retention in Telecom Using Data-Driven Segmentation and Churn Prediction

Abstract

Customer churn poses a significant challenge in the telecommunications industry, where retaining existing customers is often more cost-effective than acquiring new ones. Accurate churn prediction and meaningful customer segmentation are therefore essential for designing effective retention strategies. In this project, we present an integrated approach that combines unsupervised customer segmentation with supervised churn prediction using the Telco Customer Churn dataset.

We first perform comprehensive data preprocessing, including cleaning invalid records, encoding categorical variables, standardizing numerical features, and carefully separating the churn label to prevent data leakage. We then apply K-Means clustering to identify distinct customer segments, selecting the optimal number of clusters using the Elbow Method and Silhouette Score. The resulting clusters reveal clear differences in customer behavior, contract type, and churn tendencies. Building on these insights, we train and evaluate multiple supervised learning models for churn prediction, including logistic regression and tree-based methods, using stratified train-test splits and standard evaluation metrics.

Our results demonstrate that integrating customer segmentation with churn prediction improves interpretability and provides actionable insights into high-risk customer groups. This combined framework highlights the value of leveraging both unsupervised and supervised learning techniques for effective churn management in the telecommunications domain.

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1 Introduction

Customer churn is a critical challenge in the telecommunications industry, where acquiring new customers is often significantly more expensive than retaining existing ones. With increasingly competitive markets and low switching costs, telecom providers must proactively identify customers who are at risk of churning and design targeted retention strategies. As a result, accurate churn prediction and meaningful customer segmentation have become essential components of data-driven decision-making in customer relationship management.

Traditional churn prediction approaches primarily rely on supervised learning models that classify customers as churners or non-churners based on historical data. While such models can achieve strong predictive performance, they often provide limited insight into why customers churn and how different customer groups exhibit distinct behavioral patterns. Customer segmentation through unsupervised learning offers a complementary perspective by uncovering latent structures in customer data, enabling businesses to understand heterogeneous customer behaviors and tailor retention strategies accordingly.

In this project, we propose an integrated framework that combines customer segmentation and supervised churn prediction using the Telco Customer Churn dataset. We first perform comprehensive data preprocessing, including feature encoding, standardization, and careful separation of labels to avoid data leakage. We then apply K-Means clustering to identify meaningful customer segments and analyze their behavioral and churn characteristics. To determine the optimal number of clusters, we utilize both the Elbow Method and Silhouette Score. Finally, we build and evaluate multiple supervised learning models for churn prediction, leveraging insights from customer segmentation to enhance interpretability and business relevance. Our approach demonstrates how combining unsupervised and supervised learning techniques can provide both predictive accuracy and actionable insights for churn management in the telecommunications domain

2 Related Work

Customer churn prediction and segmentation have been widely researched in the telecommunications, e-commerce, and financial sectors. This prior research can be grouped into three main categories: (1) customer segmentation through clustering approaches, (2) predictive modeling for classifying churn, and (3) hybrid and explainable frameworks that combine both approaches.

Customer Segmentation
Unsupervised learning methods such as K-Means and hierarchical clustering have long been applied for market segmentation. (1) used an unsupervised algorithm, K-Means, combined with dimensionality reduction (PCA) to extract customer behavior patterns for targeted marketing. (2) expanded the analysis to include demographic features and spending behavior for telecom segmentation, identifying two groups: "price-sensitive" and "contract-loyal." Recent work by (3) introduced density-based clustering (DBSCAN) to better address non-linear customer distributions, demonstrating improved stability compared to K-Means method. Most studies on customer segments provide general information but do not connect these groups to the risk of losing customers or how well they predict future behavior.

Churn Prediction
Machine learning-based churn modeling has gained popularity due to its ability to identify complex relationships. (4) compared Logistic Regression, Random Forest, and XGBoost using the Telco Customer Churn dataset, achieving an accuracy of approximately 82%. (4) evaluated deep neural networks against ensemble methods, showing that simpler tree-based ensemble models perform comparably while offering higher explainability. (6) introduced profit-based evaluation metrics for churn modeling, highlighting the importance of aligning models with business value rather than focusing solely on accuracy. However, there is still limited research exploring interpretability for marketing analysts.

Hybrid and Explainable Approaches
To connect segmentation and prediction, hybrid data-mining frameworks have been developed. (7) proposed a combined clustering and classification pipeline for telecom customer retention, but it lacked visualization and interpretability. (8) examined temporal segmentation alongside churn modeling to study customer life cycles, emphasizing the importance of time-dependent patterns. (9)

introduced the SHAP framework, which enables feature-level interpretability across complex models. This approach has been widely adopted in churn analytics to identify key drivers such as tenure and contract type (10; 11). Despite these advancements, few studies integrate segmentation, predictive modeling, and explainability into a unified and interpretable framework.

Our Contribution
Building on these studies, our project proposes an integrated approach that includes: (1) using clustering to identify behavior-based customer segments, (2) applying supervised learning for churn prediction, and (3) leveraging SHAP for interpretability. Unlike previous efforts that focused on either clustering or classification in isolation, we emphasize a comprehensive data-mining workflow and the visualization of actionable business insights.

3 Methodology

3.1 Data Preprocessing

Before applying both unsupervised and supervised learning models, a unified data preprocessing pipeline was applied to ensure data quality and model compatibility across the entire project. **Binary Mapping** Categorical features with two categories (e.g., gender, Partner, Dependents, PhoneService, PaperlessBilling) were converted into binary numerical variables (0/1), allowing the models to process these features directly during training.

One-Hot Encoding All remaining multi-category categorical features (e.g., InternetService, Contract, PaymentMethod) were transformed using one-hot encoding to ensure compatibility with both linear and tree-based models.

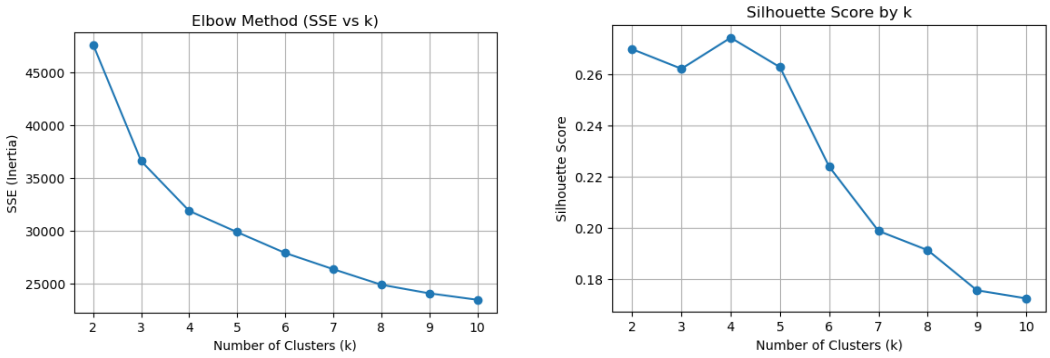
Feature Scaling Numerical features were standardized to zero mean and unit variance where appropriate, which is particularly important for distance-based methods and linear classifiers.

Train–Test Split For supervised churn prediction, the dataset was split into training and testing sets using an 80/20 ratio. Stratified sampling was applied to preserve the original class distribution of churn and non-churn customers, mitigating evaluation bias caused by class imbalance.

To prevent data leakage, the churn label was excluded from the feature set during customer segmentation.

3.2 Customer Segmentation via K-Means

To capture latent behavioral patterns among customers, we applied unsupervised clustering using the K-Means algorithm. Clustering was performed on the preprocessed feature set after encoding categorical variables and standardizing numerical features. The churn label was excluded during clustering to avoid label leakage. To determine the optimal number of clusters, we evaluated values of k ranging from 2 to 10 using both the Elbow Method and Silhouette Score. Based on these metrics, we selected $k = 4$ as a balance between cluster compactness and separation.



(a) Elbow Method (SSE vs. k)

(b) Silhouette Score vs. k

Figure 1: Selection of the number of clusters using Elbow Method and Silhouette Score.

For visualization purposes, Principal Component Analysis (PCA) was applied to project the high-dimensional feature space into two dimensions. This projection was used solely for visualization and did not affect the clustering process.

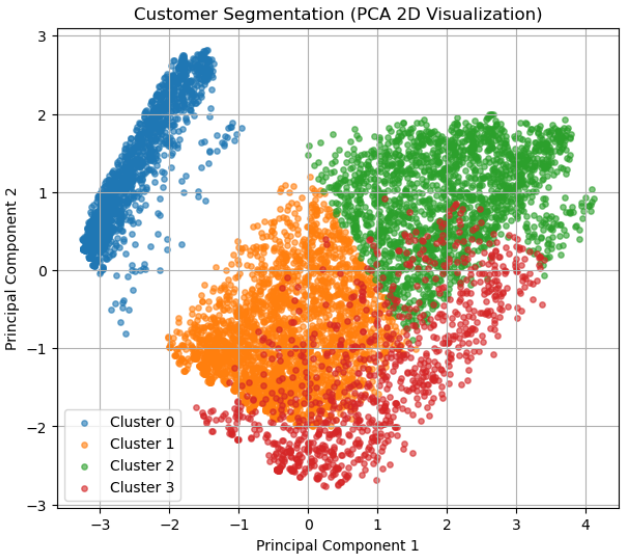


Figure 2: PCA-based 2D visualization of customer clusters obtained using K-Means.

3.3 Supervised Churn Prediction Models

Following customer segmentation, supervised learning models were developed to predict customer churn. The objective of this stage was not only to achieve strong classification performance, but also to compare different models in terms of interpretability and robustness.

We trained and evaluated multiple supervised classification models, including logistic regression and tree-based methods. Logistic regression was selected as a baseline due to its simplicity and interpretability, while tree-based models were included to capture non-linear relationships between features.

Model evaluation was performed using standard classification metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. Stratified train-test splits were used to ensure fair evaluation under class imbalance conditions

3.3.1 Data Visualization

To understand the dataset, histograms comparing different features was prepared. This comparison helped to understand whether the data were imbalanced and if the various features depended on each feature.

Through this, we observed that 26.6

Figure 3 shows the churn distribution relative to the contract type which had highest importance in model prediction.

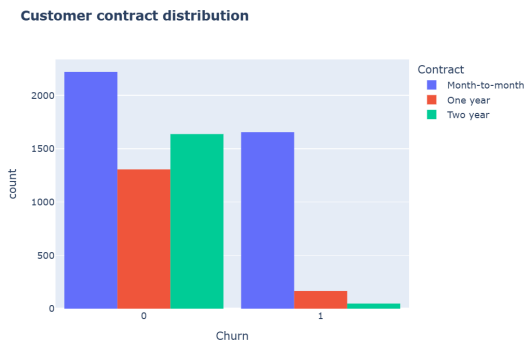


Figure 3: Churn Distribution wrt Contract type.

3.3.2 Model Selection

Given that the response variable is binary, we selected three commonly used classification models that differ in complexity and interpretability: Logistic Regression, Random Forest, and XGBoost.

Logistic Regression Logistic Regression was used as a baseline model. It is widely applied in binary classification tasks due to its simplicity and interpretability. By modeling the log-odds of churn as a linear combination of features, Logistic Regression provides a clear reference for comparing more complex models.

To estimate parameters, maximum likelihood estimation is used to maximize the likelihood that the predicted conditional probability of the target matches the observed conditional probability in the data, where the target is modeled as a sigmoid function with values between 0 and 1.

The formula is shown as follows:

$$\pi(x) = P(Y = 1 \mid X) = \frac{e^{\alpha+\beta_1 x_1+\cdots+\beta_p x_p}}{1 + e^{\alpha+\beta_1 x_1+\cdots+\beta_p x_p}}$$

Applying the logit transformation, the model can be expressed in terms of log-odds:

$$logit(\pi(x)) = \log\left(\frac{P(Y = 1 \mid X)}{1 - P(Y = 1 \mid X)}\right) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p$$

Random Forest Classifier Random Forest can capture potential nonlinear relationships between customer attributes and churn behavior. As an ensemble of decision trees built via bootstrap sampling, Random Forest reduces variance compared to a single decision tree and is generally more robust to data noise. In addition, feature importance scores from Random Forest provide useful insights into which variables contribute most to churn prediction.

XGBoost Classifier XGBoost is an extension of Gradient Boosted Decision Trees that improves model performance through regularization and an optimized tree-building process. It trains trees sequentially using an additive approach, in which each new tree corrects the errors of previous trees. During training, a greedy algorithm selects splits based on their contribution to reducing the objective function, while regularization helps prevent overfitting.

The objective function of XGBoost is defined as:

$$Objective = \sum_{i=1}^{n} L\big(y_i, \hat{f}(x_i)\big) + \sum_{m=1}^{M} \Omega(h_m)$$

where $L\big(y_i, \hat{f}(x_i)\big)$ is the loss function, and $\Omega(h_m)$ is the penalization term. Model hyperparameters for Random Forest and XGBoost were manually tuned to avoid overfitting while still improving performance.

4 Evaluation

4.1 Evaluation Metrics

This section evaluates the performance of the churn prediction models using the same testing dataset. To assess model performance, multiple evaluation metrics were considered, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy alone is insufficient for this task because the dataset is imbalanced, with non-churn customers comprising the majority class. Therefore, Recall and F1-score were emphasized to evaluate the models' ability to correctly identify churn customers. ROC-AUC was also used to assess the overall discriminative ability of each model across different classification thresholds.

In addition to numerical metrics, ROC curves and confusion matrices were generated to visualize model performance and to better understand the trade-offs between false positives and false negatives.

4.2 Churn Prediction Results

Table 1 summarizes the predictive performance of Logistic Regression, Random Forest, and XGBoost on the testing dataset. Overall, all three models achieved similar accuracy values of approximately 81.5%, indicating that accuracy alone is not sufficient to distinguish model performance in this churn prediction task.

Precision values are comparable across models, with XGBoost achieving the highest precision of 0.6936. This result indicates that when XGBoost predicts a customer as a churner, approximately 69% of such predictions are correct. Based on Recall, Logistic Regression performs best with a value of 0.5588, followed closely by XGBoost at 0.5508, while Random Forest has the lowest Recall at 0.5321, indicating that it fails to identify a larger portion of true churn customers.

The F1-score reflects a similar pattern. Logistic Regression achieved the highest F1-score of 0.6174, suggesting a better balance between Precision and Recall. These results indicate that although tree-based models are capable of capturing nonlinear patterns, the linear baseline remains competitive when considering both false positives and false negatives.

XGBoost achieved the highest ROC-AUC score of 0.8656, slightly outperforming the other models. This indicates that XGBoost has a stronger overall ability to distinguish between churn and non-churn customers across different classification thresholds. However, its Accuracy and F1-score remain comparable to those of the other models.

Table 1: Performance comparison of churn prediction models on the test dataset.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8159	0.6898	0.5588	0.6174	0.8607
Random Forest	0.8124	0.6910	0.5321	0.6012	0.8617
XGBoost	0.8159	0.6936	0.5508	0.6140	0.8656

Figure 4 presents the ROC curves for all three models. The curves are very similar and remain well above the diagonal reference line, indicating that each model can effectively distinguish between churn and non-churn customers. Although XGBoost achieves the highest ROC-AUC value, the visual differences among the curves are minimal, which is consistent with the results reported in Table 1.

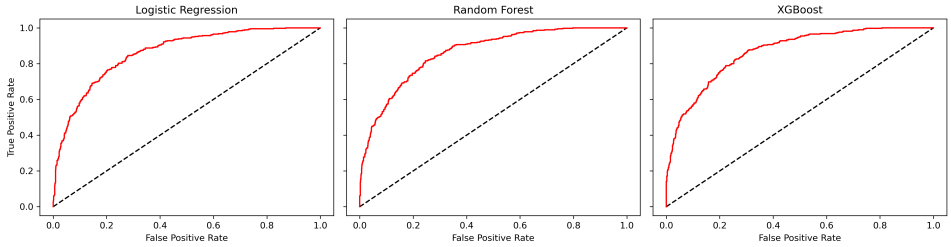


Figure 4: ROC curves for Logistic Regression, Random Forest, and XGBoost models on the test dataset.

Figure 5 shows the confusion matrices for the three models. Logistic Regression correctly identifies more churn customers, resulting in higher Recall, while Random Forest predicts fewer false positives but misses more true churn customers. XGBoost provides a balance between these two behaviors.

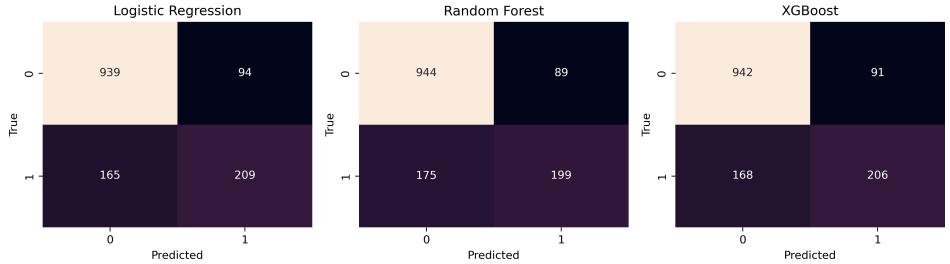


Figure 5: Confusion matrices for Logistic Regression, Random Forest, and XGBoost models.

4.3 Feature Importance Analysis

To further understand how the tree-based models make predictions, feature importance scores (Table 2) were extracted from Random Forest and XGBoost.

For Random Forest, the most important features are tenure, TotalCharges, and MonthlyCharges. Together, these variables describe the length and intensity of a customer's relationship with the company. The high importance of tenure indicates that long-term customers are less likely to churn. Similarly, TotalCharges and MonthlyCharges show that customers' total and ongoing expenses significantly influence churn behavior. Additionally, service-related features like InternetService_Fiber optic and contract-related variables rank highly, indicating the influence of service type and contract commitment.

In contrast, XGBoost focuses on categorical features for services and contracts. This indicates that XGBoost is more sensitive to structural differences in customer plans rather than overall usage patterns. Features like PaymentMethod_Electronic check and OnlineSecurity options further show that billing behavior and service security are key factors in predicting churn.

Despite these differences, both models consistently identify contract length, internet service type, and billing-related variables as key predictors of customer churn. This strengthens confidence in the model results and improves their interpretability for business insights.

Table 2: Top-10 feature importance rankings for Random Forest and XGBoost models.

Rank	Random Forest Feature	Importance	XGBoost Feature	Importance
1	tenure	0.1800	InternetService_Fiber optic	0.1673
2	TotalCharges	0.1723	Contract_Two year	0.1389
3	MonthlyCharges	0.1069	Contract_One year	0.1303
4	InternetService_Fiber optic	0.0860	InternetService_No	0.0779
5	Contract_Two year	0.0698	PaymentMethod_Electronic check	0.0662
6	PaymentMethod_Electronic check	0.0552	tenure	0.0606
7	OnlineSecurity_Yes	0.0371	OnlineSecurity_No internet service	0.0543
8	Contract_One year	0.0308	StreamingMovies_Yes	0.0304
9	TechSupport_Yes	0.0268	PaperlessBilling	0.0254
10	PaperlessBilling	0.0221	OnlineSecurity_Yes	0.0254

4.4 Model Improvement

To further improve the prediction power of the model, advanced ensemble models like Extra Trees Classifier and XGBoost with changed hyper parameter were introduced.

For better results, the model parameters were changed by first obtaining the best parameters using RandomizedSearchCV, and using those parameters to train the models. RandomizedSearchCV is a hyperparameter tuning technique which automatically tests multiple combinations of parameters and finds the best set based on a scoring metric. This technique tests combinations using random subsets and has a lower computational cost compared to GridSearchCV. Since tree based ensemble models have many hyperparameters that affect the model results, RandomizedSearchCV quickly finds a strong combination with exhaustively testing all possibilities.

Extra Trees Classifier is an advanced ensemble learning method based on decision trees. It builds multiple trees on random subsets of data, but introduces randomness by selecting split thresholds randomly rather that computing the optimal split. This often reduces overfitting and variance while still maintaining high predictive power.

Using the **Stack Classifier** to stack the most powerful models (Random Forest, XGBoost, ExtraTrees), the predictive received a significant boost. In a stacked classifier, predictions from base learners were used as input to meta-model, which learns how to best combine them. This outperformed individual models because it leveraged the strengths of each base learner, reduced biases, and captured complex patterns in the data, thus improving the evaluation metrics.

(Table 3) shows that using RandomizedSearchCV to define the hyperparameters caused a significant change in the prediction results. Stack Classifier provided the best result combining the predictive power of the three powerful models for this dataset.

Table 3: Performance comparison after hyperparameter tuning.

Model	Accuracy mean	F1-score mean	ROC-AUC mean
Random Forest	0.8351	0.8415	0.9145
XGBoost	0.8213	0.8272	0.8976
Extra Tree	0.8261	0.8314	0.9064
Stacking	0.8435	0.8459	0.9192

Figure 6 shows the ROC-AUC curve of the best predictive model (Stack Classifier), which obtained the result by integrating the best combining features of the ensemble models.

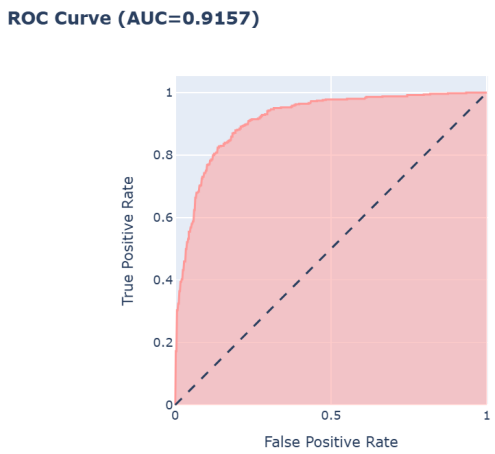


Figure 6: ROC curve for Stack Classifier (Stacking Random Forest, XGBoost, Extra Trees Classifier).

5 Discussion

This project aims to find one of the most accurate models for customer churn prediction considering the important factors that lead to customer churn. The results were examined to observe the behavior and prediction power of each algorithm.

Exploratory analysis revealed that careful observation of the dataset, it was understood that customer churn is influenced by a lot of factors including the tenure, internet service provided to the customer, dependency status, contract type and payment methods. Customers with no internet service or fiber-optic service, shorter tenure and electric check payment methods showed a higher tendency to churn.

Model interpretability was recognized using **SHAP**(SHapley Additive Explanations), which provided explanations of model behavior. SHAP summary plots visualized which features had higher influence on prediction, also illustrating that lower ranked features consistently contribute marginal yet non-negligible effects.

Although these were among the strongest features, experiments showed that training models using only these features resulted in reduced predictive performance. This suggests that customer churn is a multi factorial phenomenon, where even features of low importance contribute valuable information about customer churn. Models trained on the complete feature set showed improved ROC-AUC scores, emphasizing the benefit of feature diversity over dimensionality reduction.

Tree based ensemble models like ExtraTree Classifiers demonstrated greater performance over linear models like linear regression, achieving an ROC-AUC of approximately**0.91** after hyperparameter tuning. These improvements were mainly driven by the optimization of learning rate, class imbalance handling using **scale–pos–weight** . The results, thus, indicate that the churn predictions benefit from models that are capable of modeling complex feature interactions rather than relying on linear separability. It also shows that stacking tree based models like XGBoost, Random Forest and ExtraTree can highly improve the prediction power.

Moreover, the results obtained in this study are consistent with prior work on customer churn prediction, where ensemble based models like Random Forest and XGBoost have been to shown to outperform linear models on the Telco Customer Churn dataset (5). Additionally, the model interpretability techniques such as SHAP have been used to explain complex machine learning models by quantifying each feature contrubution and improving transparency and trust in predictive systems (9).

Overall, the results underscore the importance of balancing feature selection, model expressiveness and interpretability. While feature importance metrics identify dominant predictors, maintaining a comprehensive feature set allows models to leverage weak predictors and feature interactions, resulting in better predictive performance.

6 Limitations

This study shows that tree based ensemble models when combined using StackClassifier help in achieving a strong prediction result with imbalanced data, but this study has limitations.

First, the dataset in limited to a specific telecom provider and contains historical customer behaviour. As a result, the model's generalizability to other providers and regions, seperate set of customers may be limited.

Secondly, the churn prediction task is affected by the class imbalance, which is although mitigated by using evaluation metrics like ROC-AUC, may still have an impact in real world deployment where false negatives hold significant cost.

Finally, while tree based models provide high predictive power, they increase computational complexity and reduce interpretability compared to simpler linear models. Although SHAP was used to enhance interpretability, it may not fully capture all feature interactions.

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